

# RBA

Returns-Based Analysis

08/29/2025

# 1

Summary Statistics, MPT, Stress Simulation

Report Summary Stub: Proin bibendum sit amet purus vel vestibulum. Aliquam cursus vehicula nisl, in sagittis urna vulputate non. Fusce purus tellus, vehicula vitae orci nec, efficitur pretium ex. In vitae ornare tellus. Aenean tellus justo, lacinia pharetra eleifend sit amet, convallis ac nisl. Mauris pulvinar lorem efficitur pellentesque bibendum. Ut consequat turpis eu sem dapibus pharetra. Donec volutpat ligula a massa commodo posuere. Sed non pellentesque orci.

# Selected Track Records

|           | Number of Observations | Start Date | End Date   |
|-----------|------------------------|------------|------------|
| Fund Name |                        |            |            |
| Fund A    | 273                    | 07/31/2002 | 03/31/2025 |
| Fund B    | 279                    | 01/31/2002 | 03/31/2025 |

# Summary Statistics and MPT Full History

|                 | Fund Name               | Fund A | Fund B   |
|-----------------|-------------------------|--------|----------|
| Lookback Length | Metric Name             |        |          |
| Full History    | CAGR                    | 9.77%  | 11.06%   |
|                 | Ann.Volatility          | 9.71%  | 8.51%    |
|                 | Arithmetic Mean         | 0.82%  | 0.91%    |
|                 | Median                  | 1.0%   | 0.91%    |
|                 | Minimum                 | -10.9% | -15.94%  |
|                 | Maximum                 | 8.6%   | 11.05%   |
|                 | Max Drawdown            | 22.92% | 34.51%   |
|                 | Max Drawup              | 832.1% | 1048.98% |
|                 | Average Upside          | 2.32%  | 1.65%    |
|                 | Average Downside        | -2.27% | -1.64%   |
|                 | Ann.Volatility Upside   | 5.87%  | 5.34%    |
|                 | Ann.Volatility Downside | 6.91%  | 11.12%   |
|                 | Var (95)                | 4.4%   | 1.49%    |
|                 | cVar (5)                | 5.63%  | 5.39%    |
|                 | Skew                    | -0.48  | -2.39    |
|                 | Kurtosis                | 1.03   | 18.65    |
|                 | Batting Average         | 0.67   | 0.77     |
|                 | Calmar Ratio            | 0.43   | 0.32     |
|                 | Sharpe Ratio (rf=True)  | 0.82   | 1.09     |
|                 | Sortino Ratio (rf=True) | 1.16   | 0.83     |

**Table 1.1** Full history performance and risk statistics for selected track records. Fund A full history is 273 months from 07/31/2002 to 03/31/2025. Fund B full history is 279 months from 01/31/2002 to 03/31/2025.



# Summary Statistics and MPT Most Recent Five Years

|                 | Fund Name               | Fund A  | Fund B |
|-----------------|-------------------------|---------|--------|
| Lookback Length | Metric Name             |         |        |
| 5-Year          | CAGR                    | 13.71%  | 13.67% |
|                 | Ann.Volatility          | 11.69%  | 3.68%  |
|                 | Arithmetic Mean         | 1.13%   | 1.08%  |
|                 | Median                  | 1.4%    | 1.09%  |
|                 | Minimum                 | -6.4%   | -1.65% |
|                 | Maximum                 | 8.6%    | 3.99%  |
|                 | Max Drawdown            | 15.02%  | 3.1%   |
|                 | Max Drawup              | 101.23% | 90.08% |
|                 | Average Upside          | 2.94%   | 1.34%  |
|                 | Average Downside        | -3.09%  | -0.41% |
|                 | Ann.Volatility Upside   | 7.29%   | 3.13%  |
|                 | Ann.Volatility Downside | 4.79%   | 1.9%   |
|                 | Var (95)                | 4.11%   | 0.22%  |
|                 | cVar (5)                | 5.5%    | 1.04%  |
|                 | Skew                    | -0.13   | 0.32   |
|                 | Kurtosis                | -0.45   | 0.45   |
|                 | Batting Average         | 0.7     | 0.85   |
|                 | Calmar Ratio            | 0.91    | 4.41   |
|                 | Sharpe Ratio (rf=True)  | 0.95    | 3.01   |
|                 | Sortino Ratio (rf=True) | 2.32    | 5.83   |

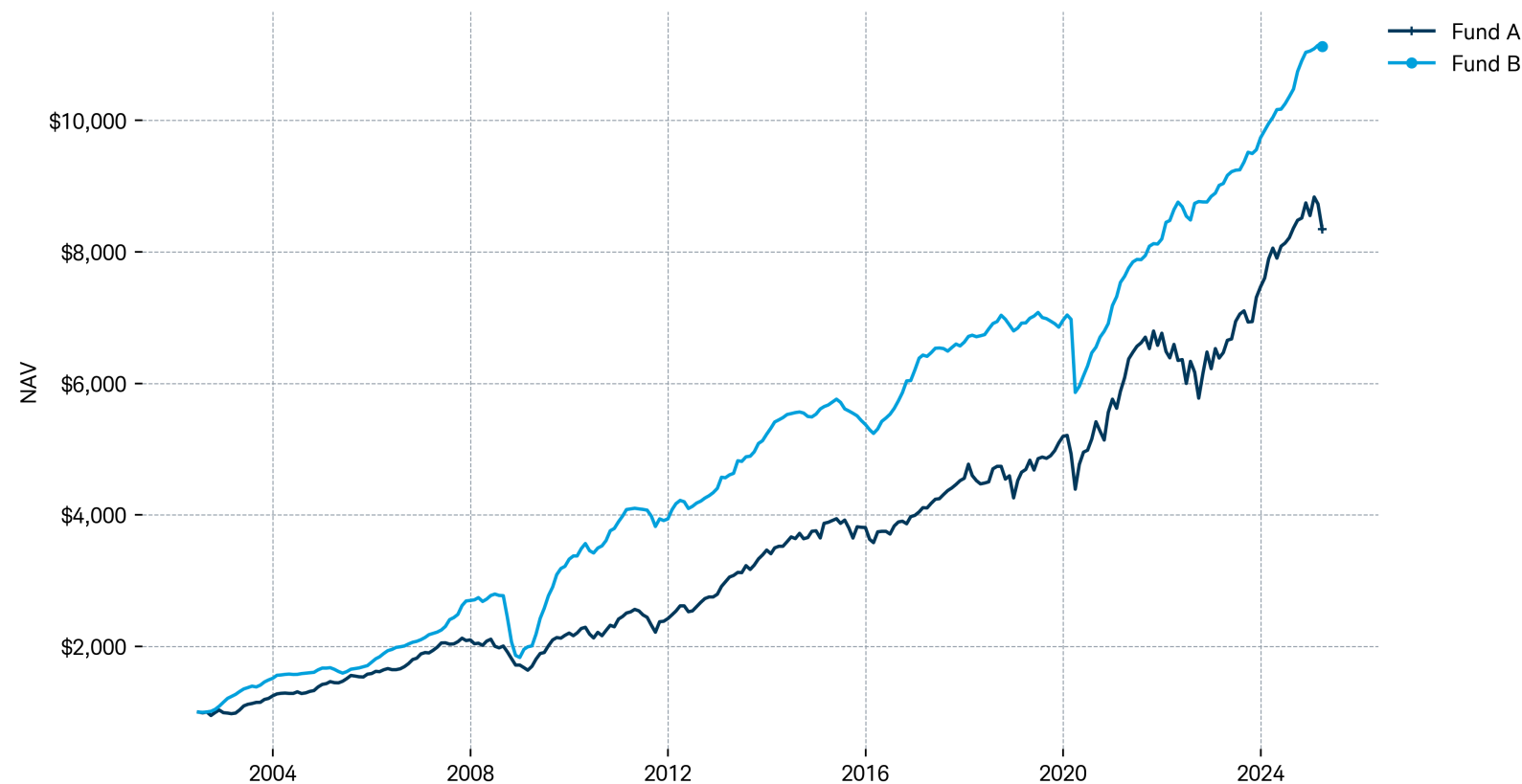
**Table 1.2** Most recent five-year performance and risk statistics for selected track records. Fund A most recent five years from 04/30/2020 to 03/31/2025. Fund B most recent five years from 04/30/2020 to 03/31/2025.

# Summary Statistics and MPT Most Recent Three Years

| Fund Name       |                         | Fund A | Fund B |
|-----------------|-------------------------|--------|--------|
| Lookback Length | Metric Name             |        |        |
| 3-Year          | CAGR                    | 8.17%  | 8.78%  |
|                 | Ann.Volatility          | 11.28% | 3.12%  |
|                 | Arithmetic Mean         | 0.71%  | 0.71%  |
|                 | Median                  | 1.15%  | 0.72%  |
|                 | Minimum                 | -6.4%  | -1.65% |
|                 | Maximum                 | 6.5%   | 2.98%  |
|                 | Max Drawdown            | 12.4%  | 3.1%   |
|                 | Max Drawup              | 52.98% | 31.32% |
|                 | Average Upside          | 2.48%  | 1.0%   |
|                 | Average Downside        | -3.33% | -0.51% |
|                 | Ann.Volatility Upside   | 6.56%  | 2.4%   |
|                 | Ann.Volatility Downside | 5.71%  | 2.04%  |
|                 | Var (95)                | 4.72%  | 0.71%  |
|                 | cVar (5)                | 6.05%  | 1.22%  |
|                 | Skew                    | -0.34  | -0.01  |
|                 | Kurtosis                | -0.45  | 1.06   |
|                 | Batting Average         | 0.69   | 0.81   |
|                 | Calmar Ratio            | 0.66   | 2.83   |
|                 | Sharpe Ratio (rf=True)  | 0.34   | 1.43   |
|                 | Sortino Ratio (rf=True) | 0.67   | 2.18   |

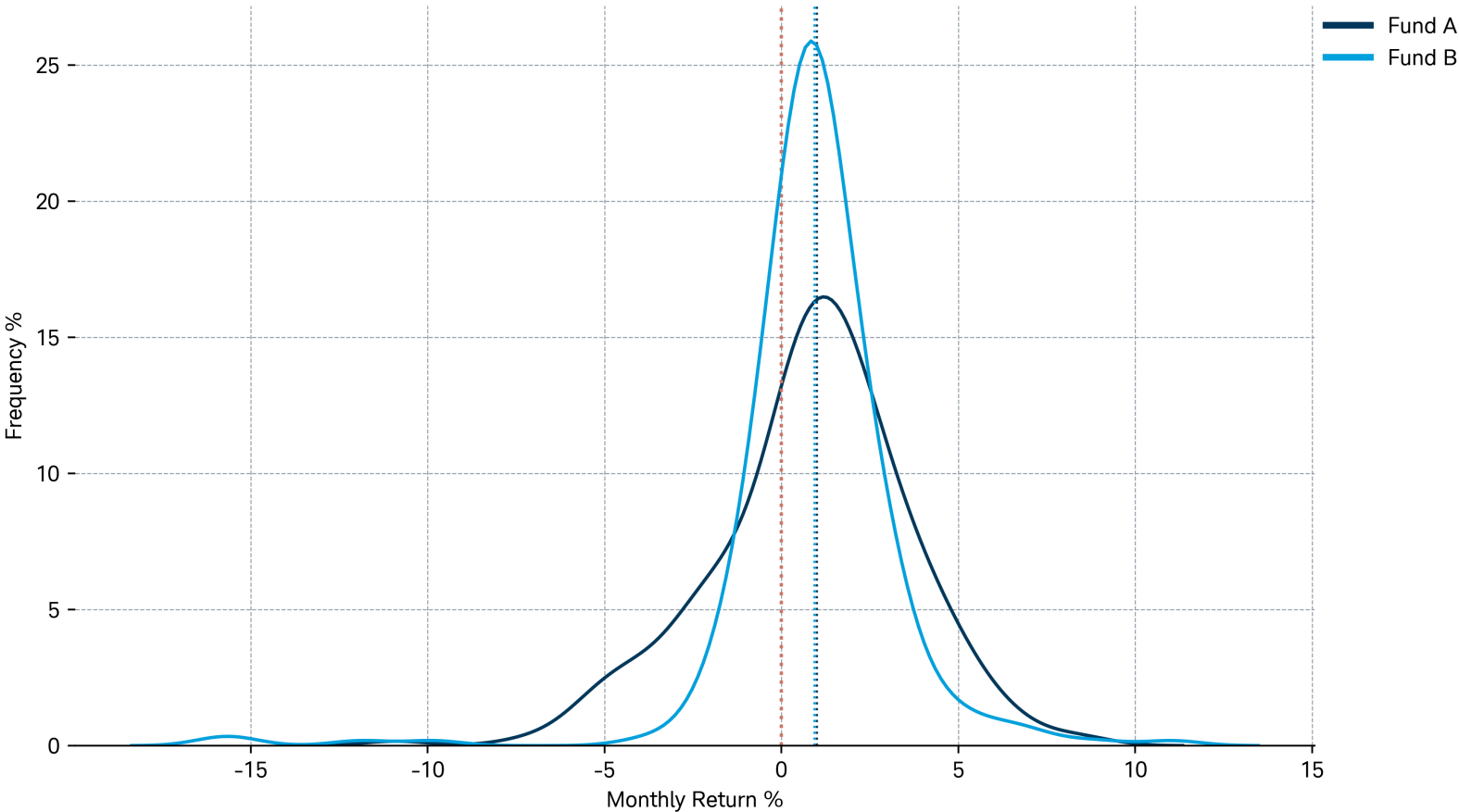
**Table 1.3** Most recent three-year performance and risk statistics for selected track records. Fund A most recent three years from 04/30/2022 to 03/31/2025. Fund B most recent three years from 04/30/2022 to 03/31/2025.

# Selected Charts Growth of \$1,000



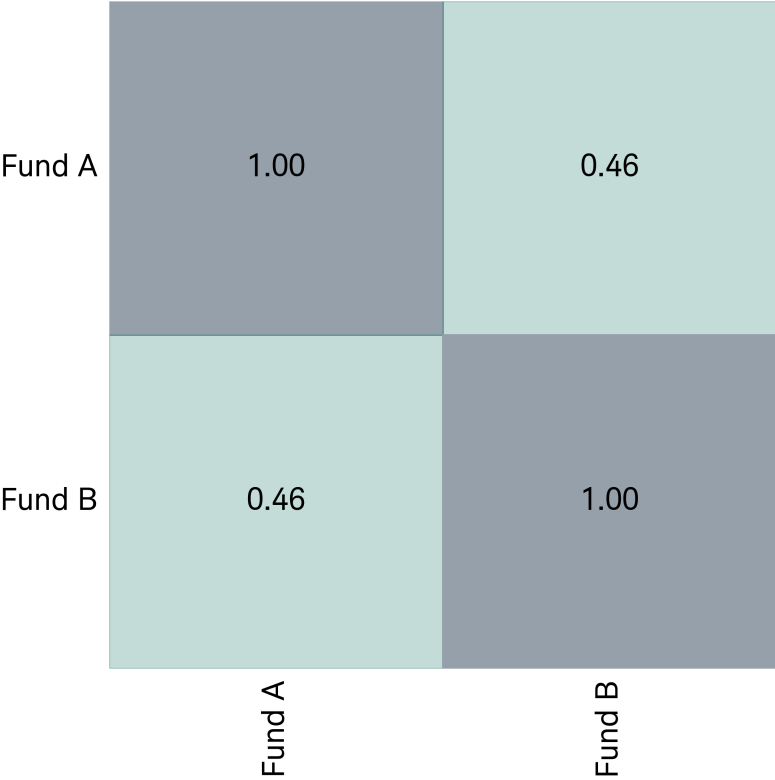
**Figure 1.1** Growth of \$1,000 for for selected track records. Intersection of selected track records is 273 months from 07/31/2002 to 03/31/2025.

# Selected Charts Distribution of Monthly Returns



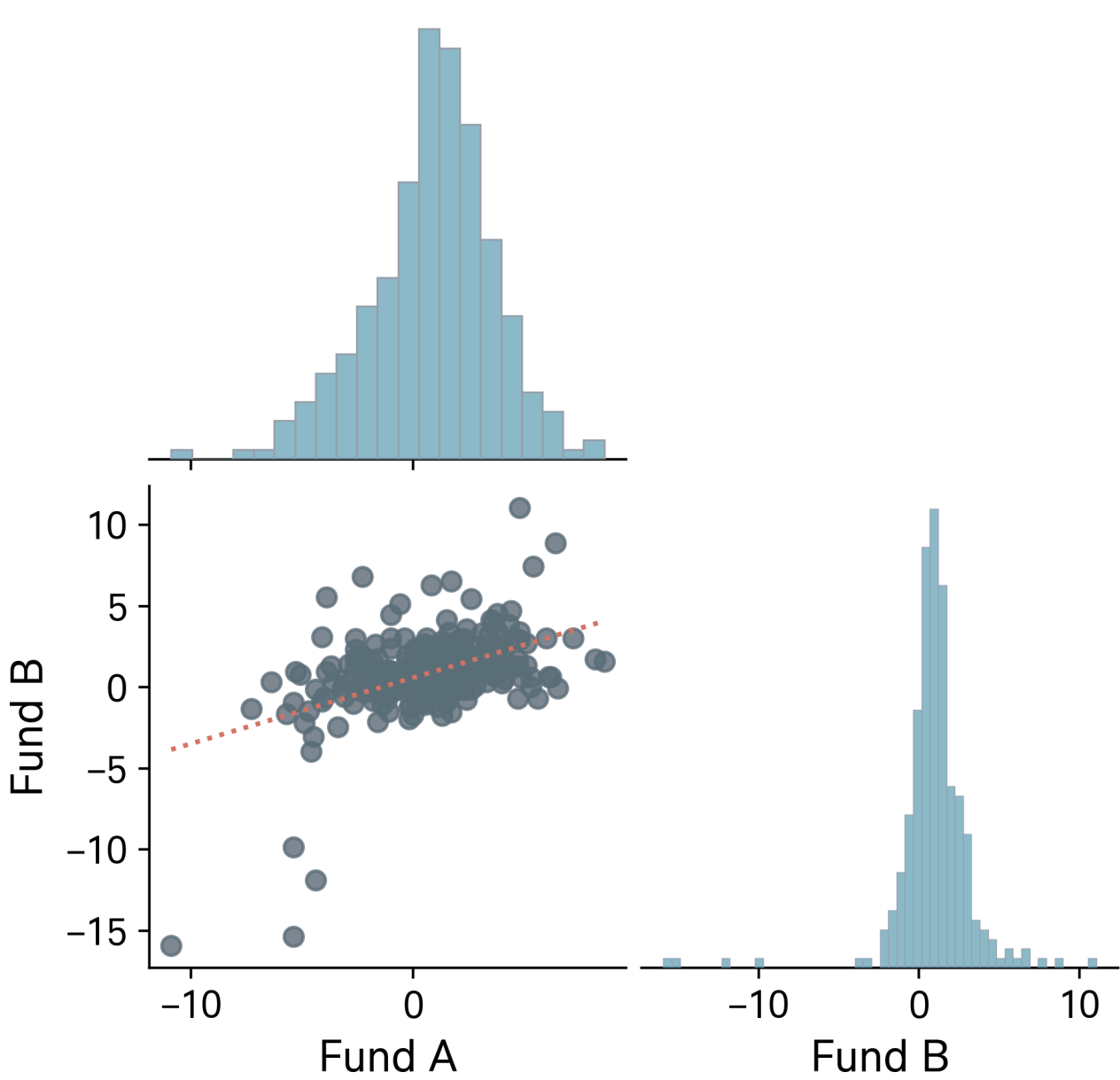
**Figure 1.2** Distribution of monthly returns for selected fund track records. Dashed lines indicate respective median. Intersection of selected track records is 273 months from 07/31/2002 to 03/31/2025.

# Selected Charts Fund Correlations



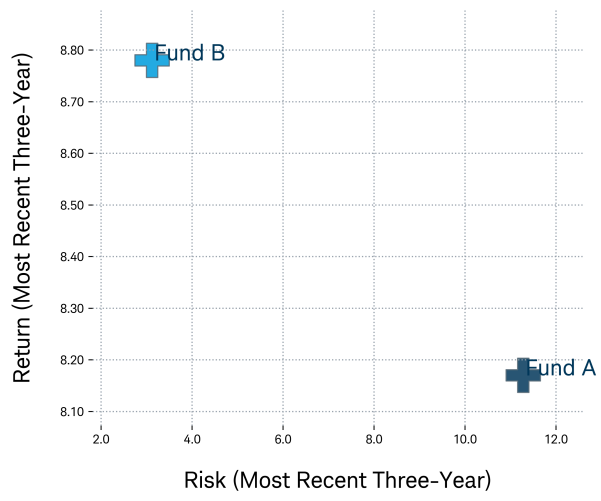
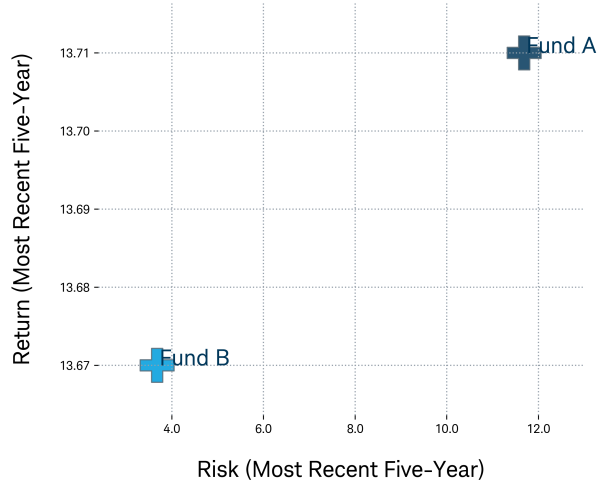
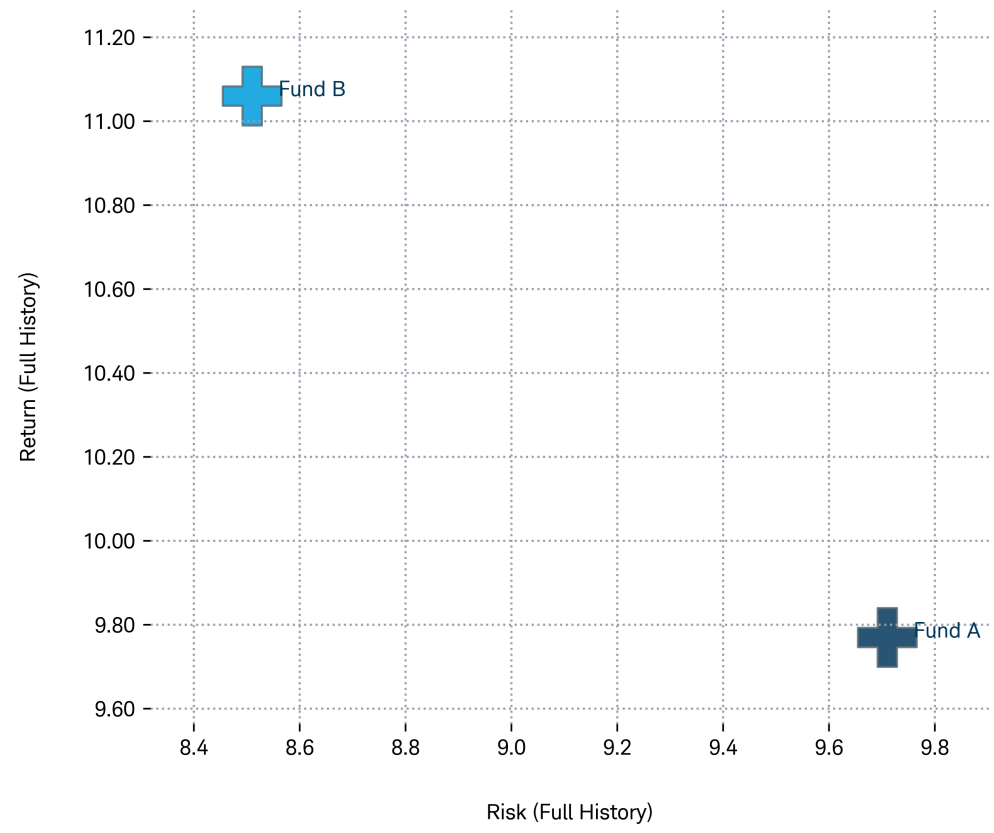
**Figure 1.3** Correlation of monthly returns for selected track records. Intersection of selected track records is 273 months from 07/31/2002 to 03/31/2025.

# Selected Charts Pairwise Scatters and Univariate Distributions



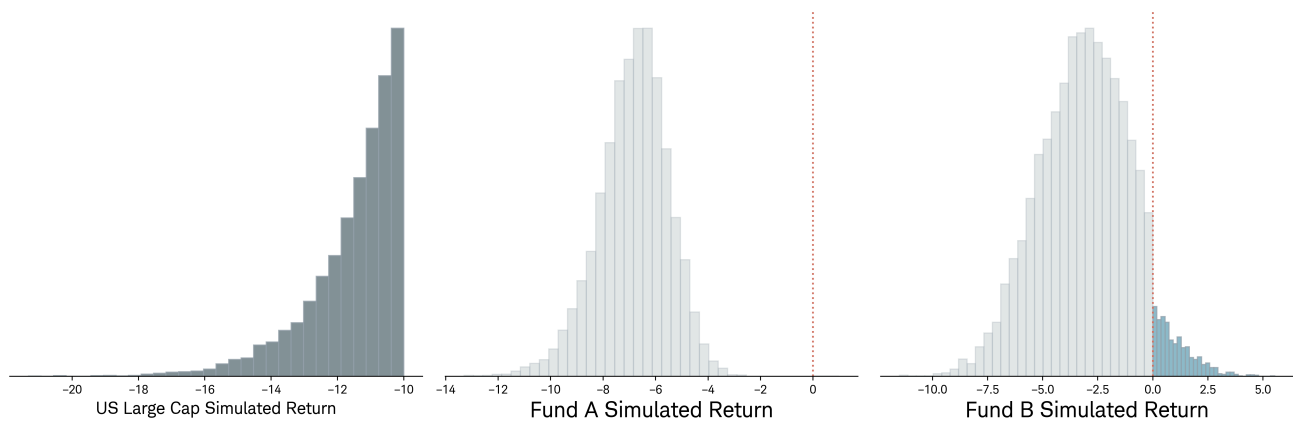
**Figure 1.4** Scatterplots of monthly returns for each pair of track records. Diagonal is the univariate distributions of monthly returns. Dashed line in scatters is the line of best fit. Intersection of selected track records is 273 months from 07/31/2002 to 03/31/2025.

# Selected Charts Risk Return Grids



**Figure 1.5** Annualized risk and return grids. Full history is intersection of selected track records, 273 months from 07/31/2002 to 03/31/2025 Most recent five-year history is 07/31/2002 to 03/31/2025. Most recent three-year history is 07/31/2002 to 03/31/2025.

# Stress Simulation US Large Cap Equity



**Table 1.4** Ranked correlations of US Large Cap and selected track records.

| Corr(US Large Cap, Fund) |      |
|--------------------------|------|
| Fund A                   | 0.92 |
| Fund B                   | 0.31 |

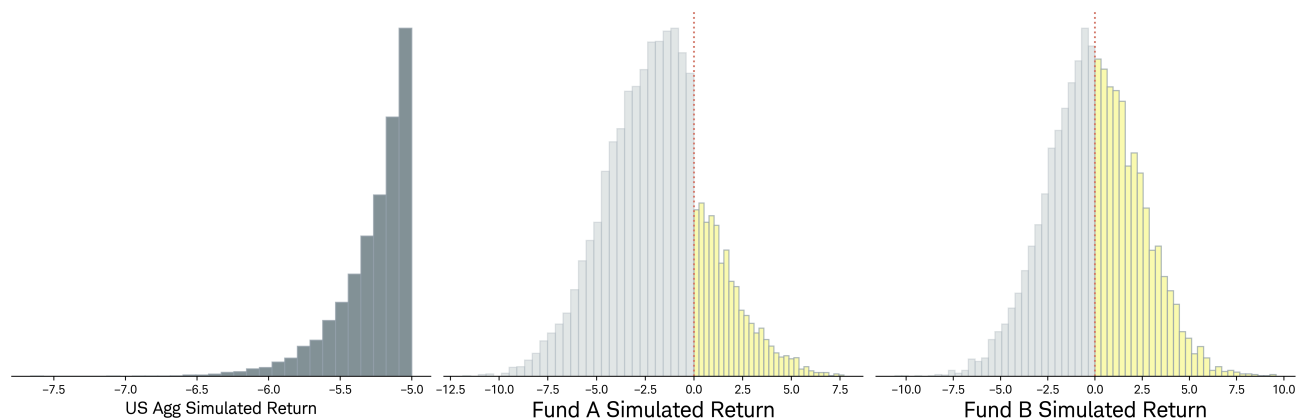
**Figure 1.6** Joint distributions of US Large Cap and selected track records when US Large Cap is stressed at least -10%.

$$\text{Acceptance Rate} = \frac{10,095}{900,000} = 1.12\%$$

Acceptance rate in stress simulation is the proportion of generated scenarios that meet the specified stress condition (e.g. market drop threshold) indicating how frequently the simulated stress event occurs within the total number of trials.



# Stress Simulation US Aggregate Bond



**Table 1.5** Ranked correlations of US Agg. and selected track records.

| Corr(US Agg, Fund) |       |
|--------------------|-------|
| Fund A             | 0.06  |
| Fund B             | -0.09 |

**Figure 1.7** Joint distributions of US Aggregate Bond and selected track records when US Aggregate Bond is stressed at least -5%.

$$\text{Acceptance Rate} = \frac{10,000}{421,950,000} = 0.0024\%$$

Acceptance rate in stress simulation is the proportion of generated scenarios that meet the specified stress condition (e.g. market drop threshold) indicating how frequently the simulated stress event occurs within the total number of trials.

# Stress Simulation Summary Tables

**Table 1.6** Summary statistics of 10,000 random correlated variables when US Large Cap is stressed at least -10%.

|              | Count  | Mean   | Median | Std.Dev. | Min.   | 25th   | 75th   | Max.   |
|--------------|--------|--------|--------|----------|--------|--------|--------|--------|
| Fund A       | 10,000 | -6.85  | -6.76  | 1.34     | -13.30 | -7.65  | -5.95  | -2.54  |
| Fund B       | 10,000 | -2.88  | -2.89  | 2.25     | -11.50 | -4.40  | -1.35  | 5.53   |
| US Large Cap | 10,000 | -11.53 | -11.13 | 1.40     | -21.32 | -12.16 | -10.50 | -10.00 |

**Table 1.7** Summary statistics of 10,000 random correlated variables when US Agg. is stressed at least -5%.

|              | Count  | Mean  | Median | Std.Dev. | Min.   | 25th  | 75th  | Max.  |
|--------------|--------|-------|--------|----------|--------|-------|-------|-------|
| Fund A       | 10,000 | -1.57 | -1.56  | 2.78     | -11.86 | -3.46 | 0.31  | 7.70  |
| Fund B       | 10,000 | -0.00 | 0.01   | 2.48     | -10.59 | -1.64 | 1.65  | 9.59  |
| US Aggregate | 10,000 | -5.27 | -5.19  | 0.26     | -7.66  | -5.39 | -5.08 | -5.00 |

# 2

Factor Loadings and Related Metrics

# Full History Comparison Factor Loadings (Beta Estimates)

|                       | Fund A | Fund B |
|-----------------------|--------|--------|
| Beta Estimate         |        |        |
| Macro Commodity       | 0.10   | 0.14   |
| Macro EM              |        |        |
| Macro FX              |        |        |
| Macro Global Rates    |        | -0.20  |
| Macro Int.Dev. Equity | 0.57   | -0.19  |
| Macro US Credit       | 0.64   | 0.53   |
| Macro US Equity       | 0.53   | -0.21  |
| Macro US Inflation    | 0.77   | 0.98   |
| Strategy FX Carry     |        |        |
| Strategy Rates Carry  |        |        |
| Strategy Short Vol.   |        | 0.16   |
| Strategy Trend        | 0.07   | 0.12   |
| Style Low Beta        |        |        |
| Style Momentum        |        |        |
| Style Quality         |        | -0.08  |
| Style Size            |        |        |
| Style Value           |        |        |

**Table 2.1** Factor loadings (betas) as computed from full history of selected track records. Fund A full history is 273 months from 07/31/2002 to 03/31/2025. Fund B full history is 279 months from 01/31/2002 to 03/31/2025.

# Full History Comparison Contribution to Risk

|                       | Fund A  | Fund B  |
|-----------------------|---------|---------|
| Macro Commodity       | 7.56%   | 18.25%  |
| Macro EM              |         |         |
| Macro FX              |         |         |
| Macro Global Rates    |         | -3.39%  |
| Macro Int.Dev. Equity | 28.07%  | 1.13%   |
| Macro US Credit       | 23.80%  | 17.53%  |
| Macro US Equity       | 17.62%  | -1.56%  |
| Macro US Inflation    | 9.92%   | 18.81%  |
| Strategy FX Carry     |         |         |
| Strategy Rates Carry  |         |         |
| Strategy Short Vol.   |         | 10.59%  |
| Strategy Trend        | 0.03%   | 0.18%   |
| Style Low Beta        |         |         |
| Style Momentum        |         |         |
| Style Quality         |         | 2.95%   |
| Style Size            |         |         |
| Style Value           |         |         |
| Unexplained           | 12.99%  | 35.52%  |
| Sum                   | 100.00% | 100.00% |

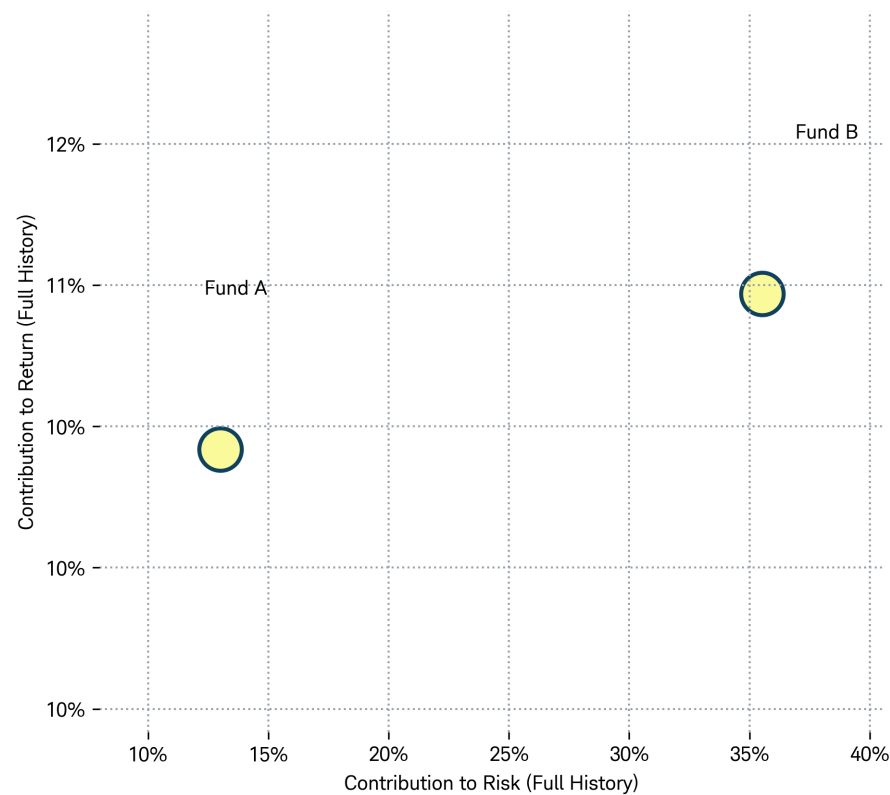
**Table 2.2** Contribution to risk as computed from full history of selected track records. Fund A full history is 273 months from 07/31/2002 to 03/31/2025. Fund B full history is 279 months from 01/31/2002 to 03/31/2025.

# Full History Comparison Contribution to Return

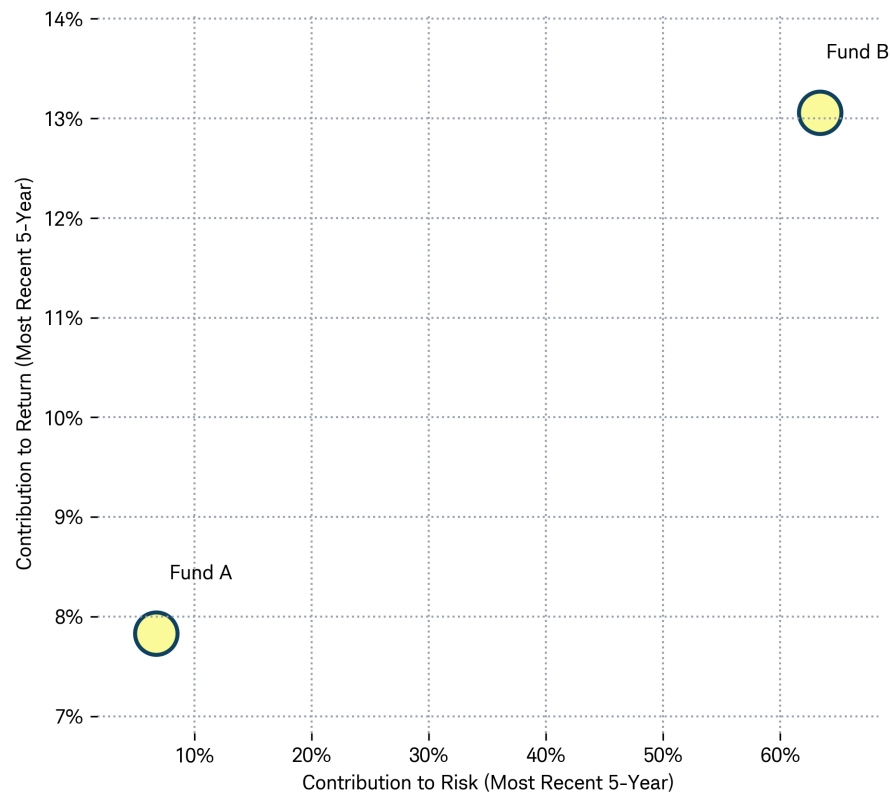
|                       | Fund A | Fund B |
|-----------------------|--------|--------|
| Macro Commodity       | -0.25% | -0.37% |
| Macro EM              |        |        |
| Macro FX              |        |        |
| Macro Global Rates    |        | -0.94% |
| Macro Int.Dev. Equity | -0.28% | 0.09%  |
| Macro US Credit       | -0.16% | -0.13% |
| Macro US Equity       | -0.16% | 0.06%  |
| Macro US Inflation    | -0.06% | -0.07% |
| Strategy FX Carry     |        |        |
| Strategy Rates Carry  |        |        |
| Strategy Short Vol.   |        | 1.14%  |
| Strategy Trend        | 0.26%  | 0.47%  |
| Style Low Beta        |        |        |
| Style Momentum        |        |        |
| Style Quality         |        | -0.15% |
| Style Size            |        |        |
| Style Value           |        |        |
| Unexplained           | 10.42% | 10.97% |
| Sum                   | 9.77%  | 11.06% |

**Table 2.3** Contribution to return as computed from full history of selected track records. Fund A full history is 273 months from 07/31/2002 to 03/31/2025. Fund B full history is 279 months from 01/31/2002 to 03/31/2025.

# Comparison Unexplained Risk/Return



**Figure 2.1** Full history of selected track records. Unexplained risk and return derived from factor contributions to risk and return. Fund A full history is 273 months from 07/31/2002 to 03/31/2025. Fund B full history is 279 months from 01/31/2002 to 03/31/2025.



**Figure 2.2** Most recent five years of selected track records. Unexplained risk and return derived from factor contributions to risk and return. Fund A most recent five years from 04/30/2020 to 03/31/2025. Fund B most recent five years from 04/30/2020 to 03/31/2025.

# Fund Analytics Full History and Rolling Windows

The following sections provide granular analysis on specific funds. Regressions and thus beta-related metrics computed from the complete track record provide a single estimate of the relationships between factors. If a track record is lengthy the regression can be underfit. Though full history data captures long-term trends it obscures important changes in behavior or risk that occur over time, especially if market conditions or fund strategies shift. Regressions on rolling windows can provide a more nuanced view. It can highlight periods of increased or decreased sensitivity, structural breaks, or regime changes, making it a more informative tool for understanding dynamic risk and return characteristics.

**For Fund A rolling windows analysis** we compute factor loadings and related metrics on 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.

**For Fund B rolling windows analysis** we compute factor loadings and related metrics on 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



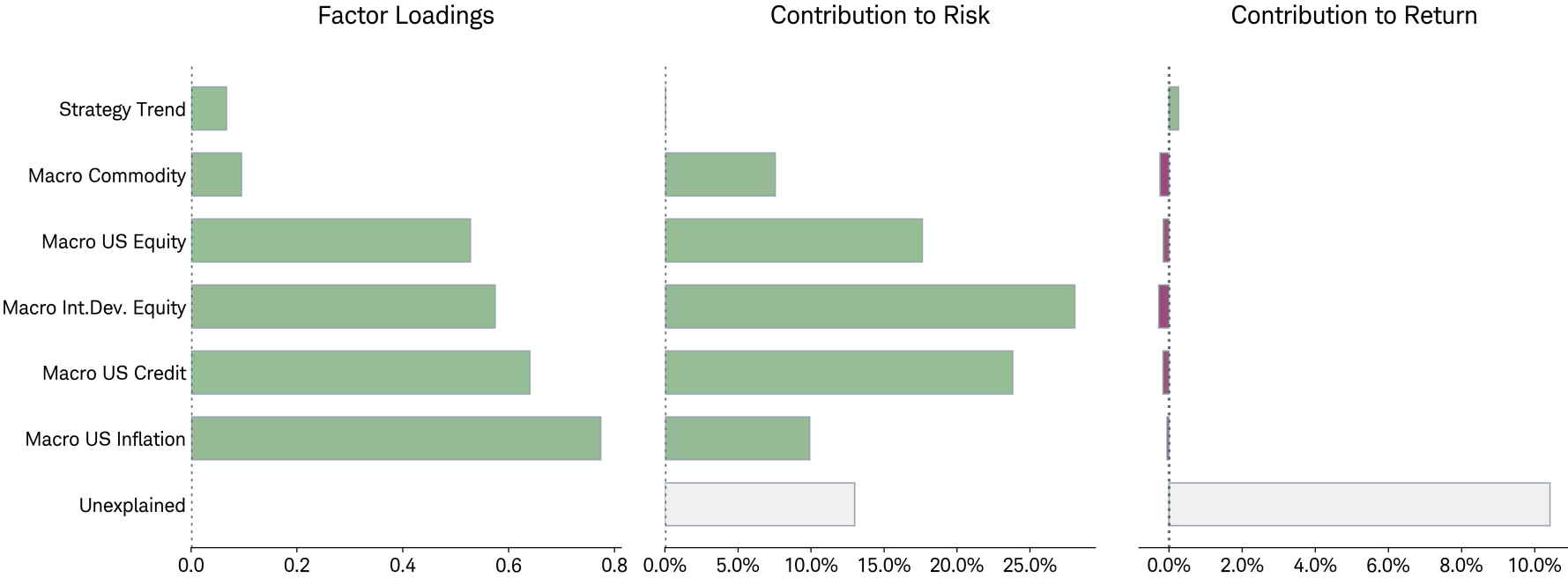
## Fund A Analytics

# Full History Factor Loadings and Related Beta Metrics

|                  |                 |             |                       | Beta Estimate | Contribution Risk | Contribution Return |
|------------------|-----------------|-------------|-----------------------|---------------|-------------------|---------------------|
| Selected Manager | Lookback Length | Return Date | Factor Name           |               |                   |                     |
| Fund A           | 273             | 03/31/2025  | Macro Commodity       | 0.10          | 7.56%             | -0.25%              |
|                  |                 |             | Macro US Inflation    | 0.77          | 9.92%             | -0.06%              |
|                  |                 |             | Macro US Credit       | 0.64          | 23.80%            | -0.16%              |
|                  |                 |             | Macro Int.Dev. Equity | 0.57          | 28.07%            | -0.28%              |
|                  |                 |             | Macro US Equity       | 0.53          | 17.62%            | -0.16%              |
|                  |                 |             | Strategy Trend        | 0.07          | 0.03%             | 0.26%               |
|                  |                 |             | Unexplained           |               | 12.99%            | 10.42%              |
|                  |                 |             | Sum                   |               | 100.00%           | 9.77%               |

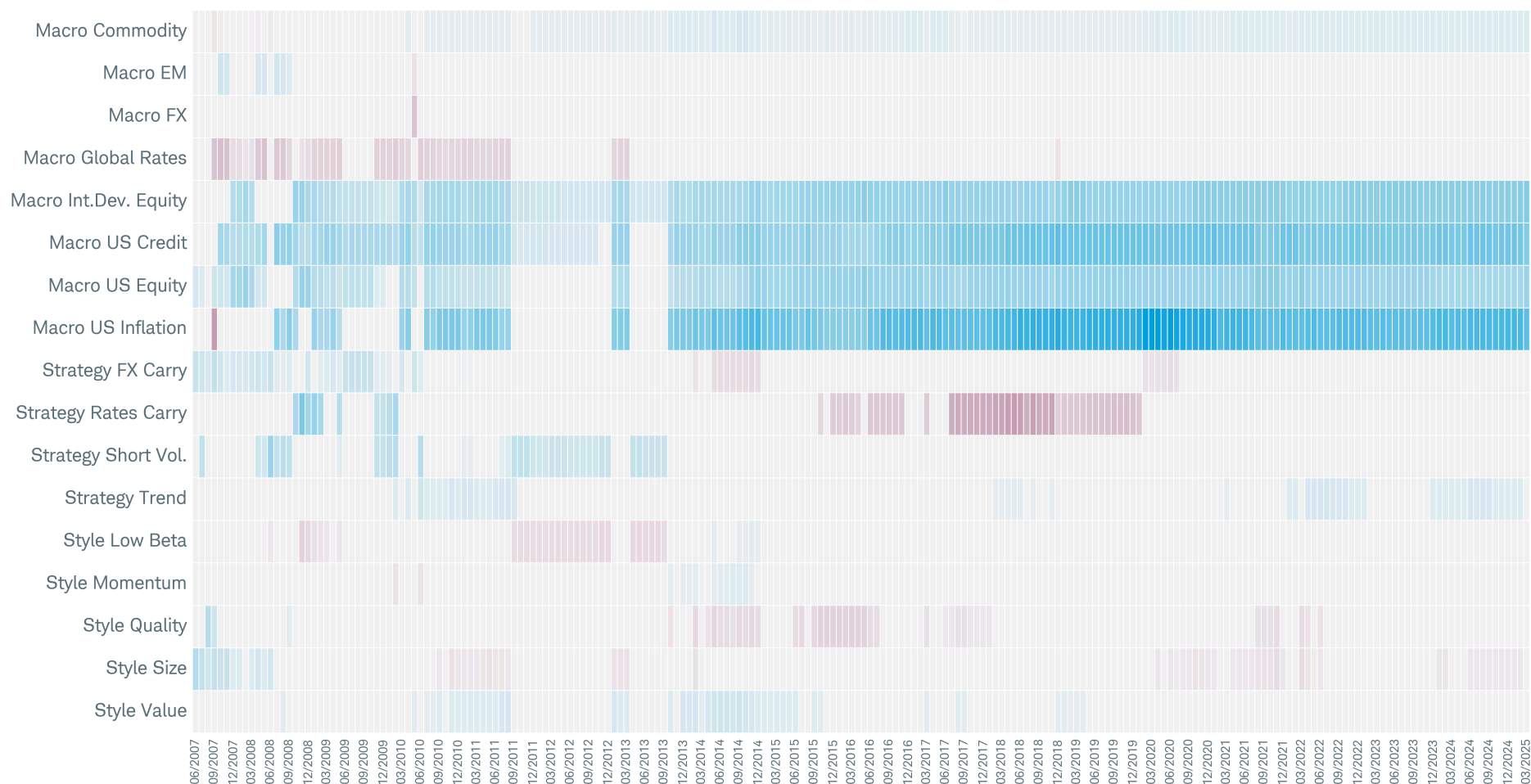
**Table 2.5** Summary of Fund A factor loadings, contribution to risk, and contribution to return. As computed from manager’s full history of 273 months ending 03/31/2025.

# Full History Factor Loadings and Related Beta Metrics

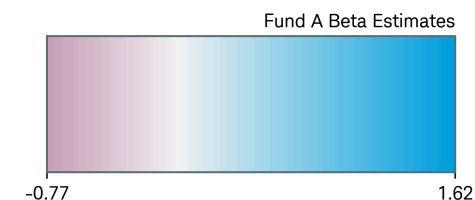


**Figure 2.4** Summary of Fund A factor loadings, contribution to risk, and contribution to return. As computed from manager’s full history of 273 months ending 03/31/2025.

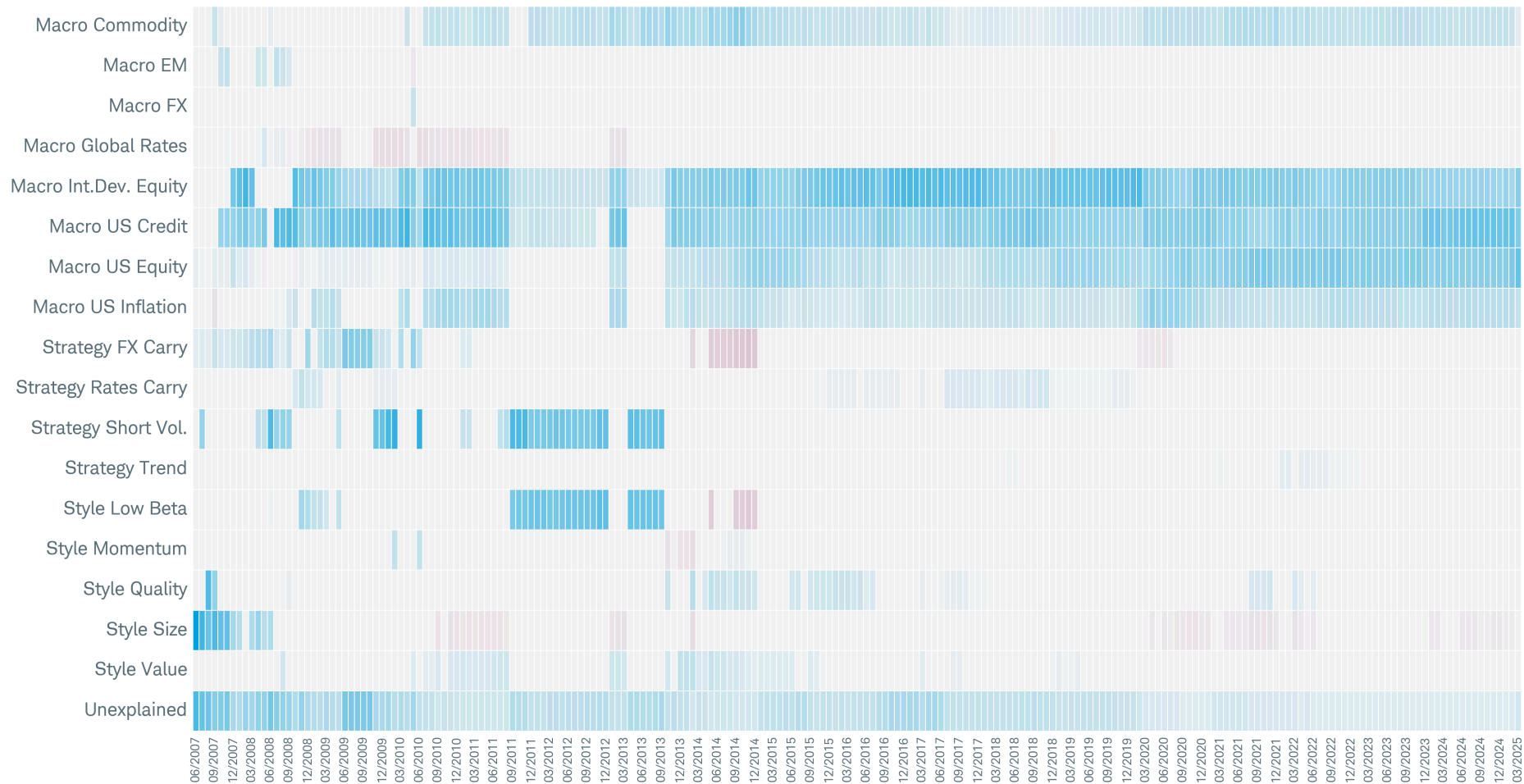
# Rolling Windows Factor Loadings (Beta Estimates)



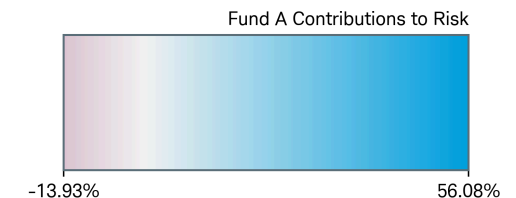
**Figure 2.5** Fund A heatmap of factor loadings (beta estimates) for 60-month rolling windows across manager's track record. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.



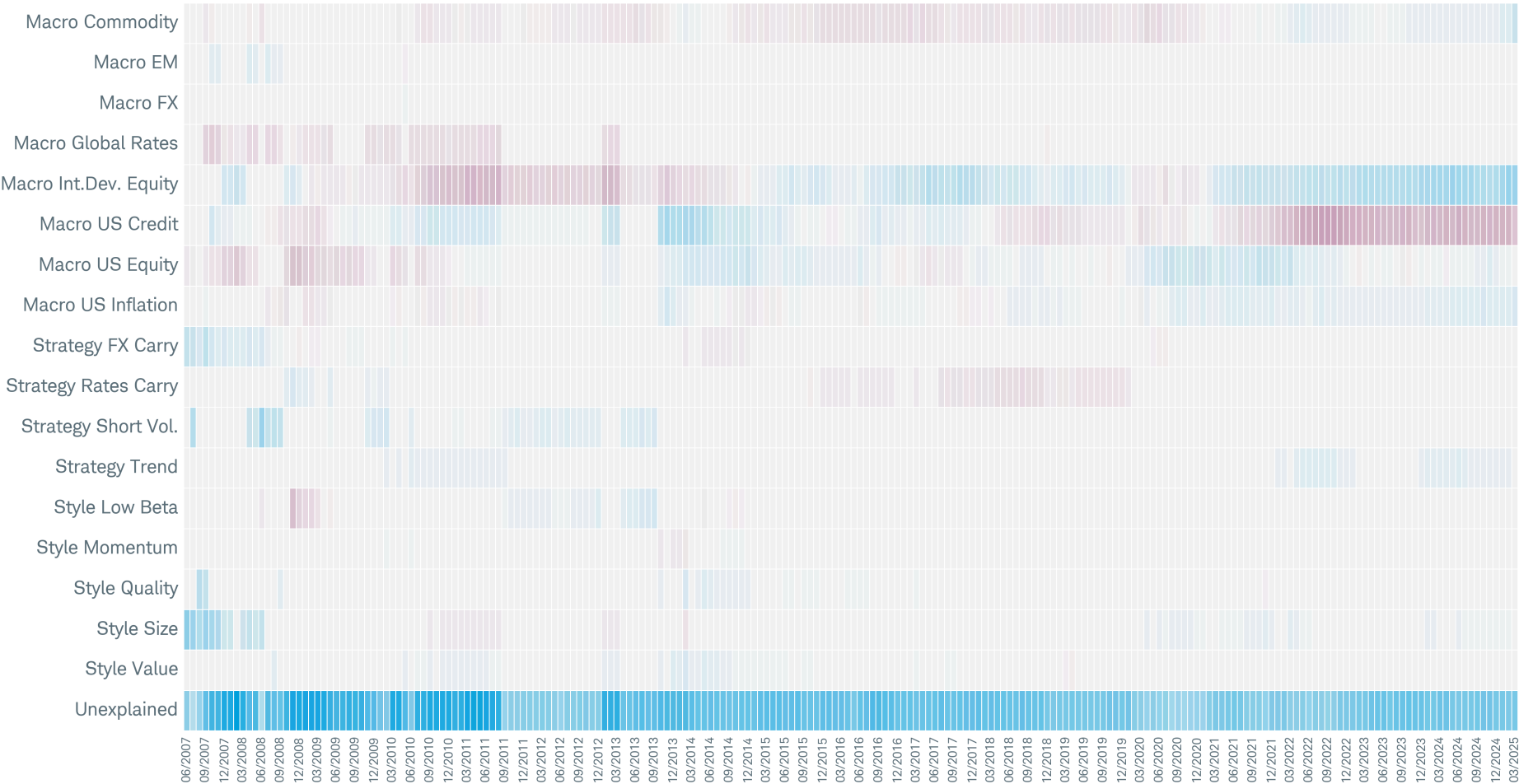
# Rolling Windows Contribution to Risk



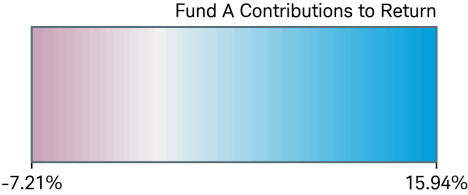
**Figure 2.6** Fund A heatmap of contribution to risk (derived from factor loadings) for 60-month rolling windows across manager's track record. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.



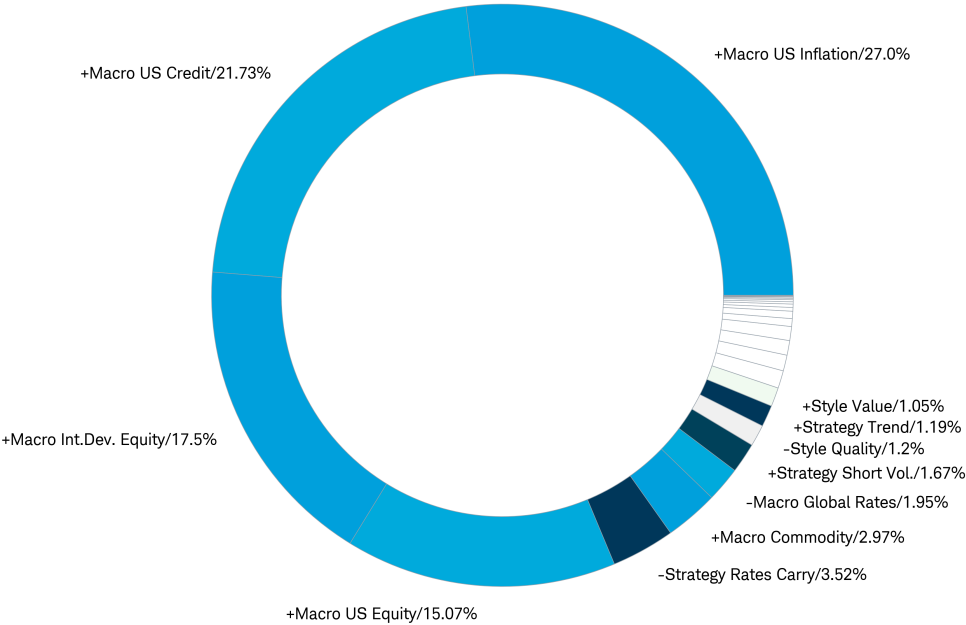
# Rolling Windows Contribution to Return



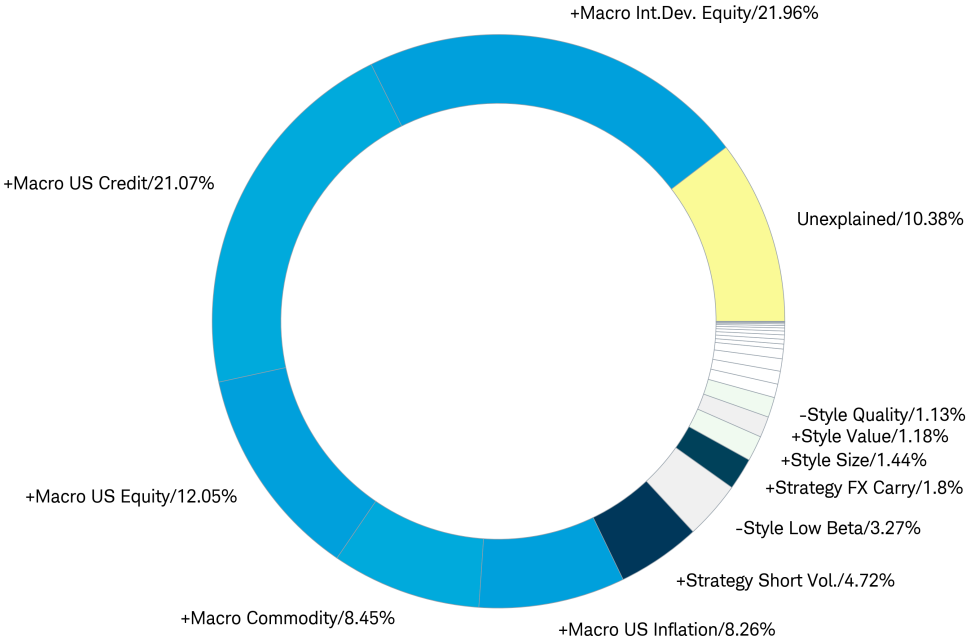
**Figure 2.7** Fund A heatmap of contribution to return (derived from factor loadings) for 60-month rolling windows across manager's track record. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.



# Rolling Windows Factor Loadings and Contribution to Risk Summary

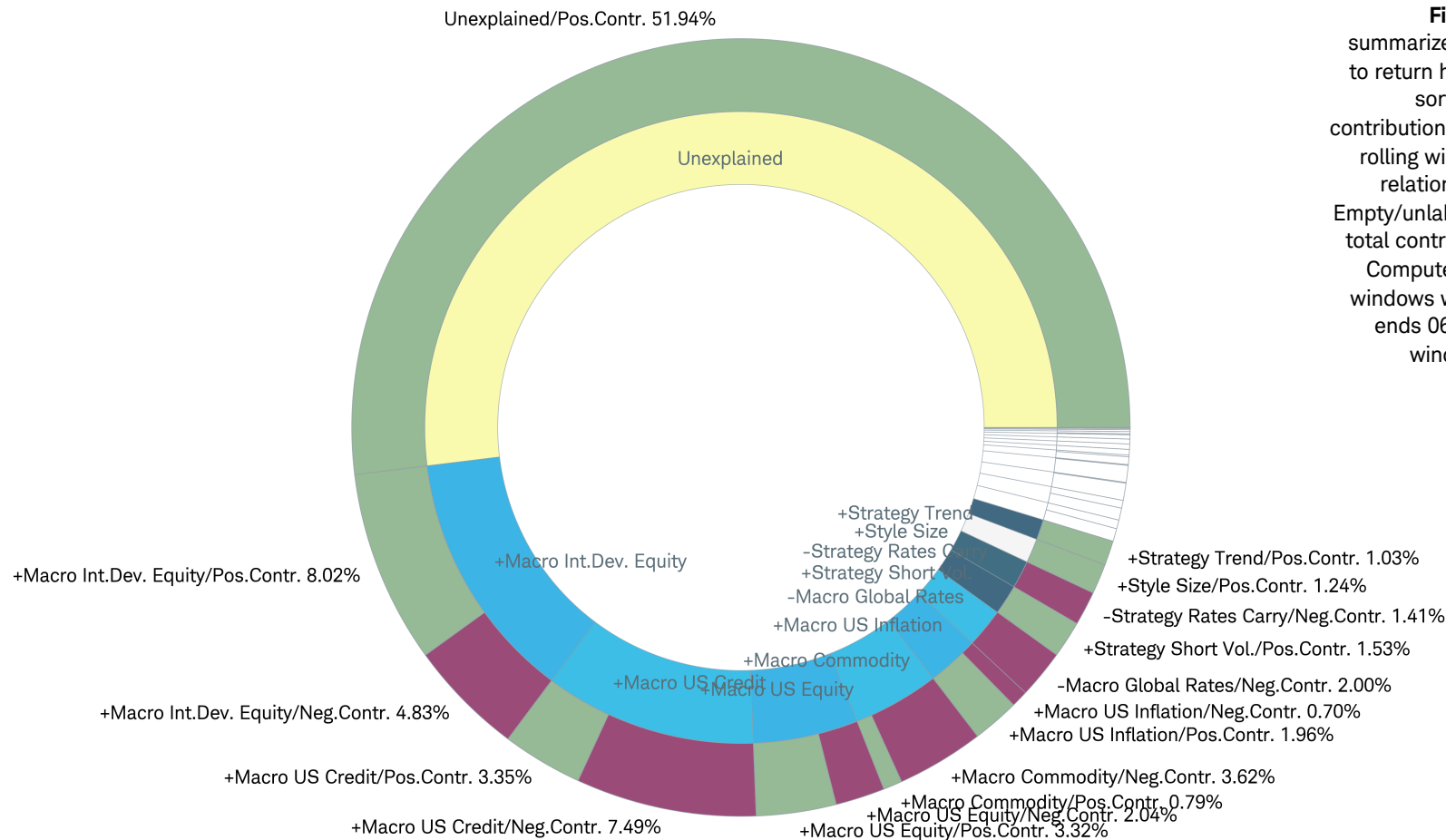


**Figure 2.8** Donut chart summarizes Fund A factor loading heatmap. Attribution is sorted by total percent of factor loadings across all rolling windows. Positive or negative beta indicated by +/- . Empty/unlabeled wedges indicate total loadings less than 1% of all loadings computed. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.



**Figure 2.9** Donut chart summarizes Fund A contribution to risk heatmap. Attribution is sorted by percent of total risk computed across all rolling windows. Positive or negative prefix indicates contribution to risk associated with positive or negative beta. Empty/unlabeled wedges indicate a factor loading associated with less than 1% of all contribution to risk. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.

## Rolling Windows Contribution to Return Summary



**Figure 2.10** Donut chart summarizes Fund A contribution to return heatmap. Attribution is sorted by total percent of contribution to return to across all rolling windows. Direct/indirect relationship indicated by +/- . Empty/unlabeled wedges indicate total contributions less than 1%. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.



# Rolling Windows Heatmap and Donut Summaries

|                        | Percent of Total<br>Factor Loadings | Percent of Loading<br>Positive Return | Percent of Loading<br>Negative Return | Percent of<br>All Return | Percent of Total<br>Risk |
|------------------------|-------------------------------------|---------------------------------------|---------------------------------------|--------------------------|--------------------------|
| Factor Loading         |                                     |                                       |                                       |                          |                          |
| Unexplained            |                                     | 51.94%                                | 0.00%                                 | 51.94%                   | 10.38%                   |
| +Macro Int.Dev. Equity | 17.50%                              | 8.02%                                 | 4.83%                                 | 12.85%                   | 21.96%                   |
| +Macro US Credit       | 21.73%                              | 3.35%                                 | 7.49%                                 | 10.84%                   | 21.07%                   |
| +Macro US Equity       | 15.07%                              | 3.32%                                 | 2.04%                                 | 5.36%                    | 12.05%                   |
| +Macro Commodity       | 2.97%                               | 0.79%                                 | 3.62%                                 | 4.41%                    | 8.45%                    |
| +Macro US Inflation    | 27.00%                              | 1.96%                                 | 0.70%                                 | 2.66%                    | 8.26%                    |
| +Strategy Short Vol.   | 1.67%                               | 1.53%                                 | 0.00%                                 | 1.53%                    | 4.72%                    |
| -Style Low Beta        | 0.88%                               | 0.53%                                 | 0.37%                                 | 0.90%                    | 3.27%                    |
| +Strategy FX Carry     | 0.98%                               | 0.71%                                 | 0.04%                                 | 0.75%                    | 1.80%                    |
| +Style Size            | 0.43%                               | 1.24%                                 | 0.00%                                 | 1.24%                    | 1.44%                    |
| +Style Value           | 1.05%                               | 0.70%                                 | 0.03%                                 | 0.73%                    | 1.18%                    |
| -Style Quality         | 1.20%                               | 0.32%                                 | 0.07%                                 | 0.39%                    | 1.13%                    |

**Table 2.6** Summary of Fund A factor loadings, contribution to risk, and contribution to return donut charts. Filtered for least 1% of risk. Percentages indicate the percent of all factor loadings, risk, and return computed across all rolling windows. Computed from 214 60-month windows where the first window ends 06/30/2007 and the last window ends 03/31/2025.

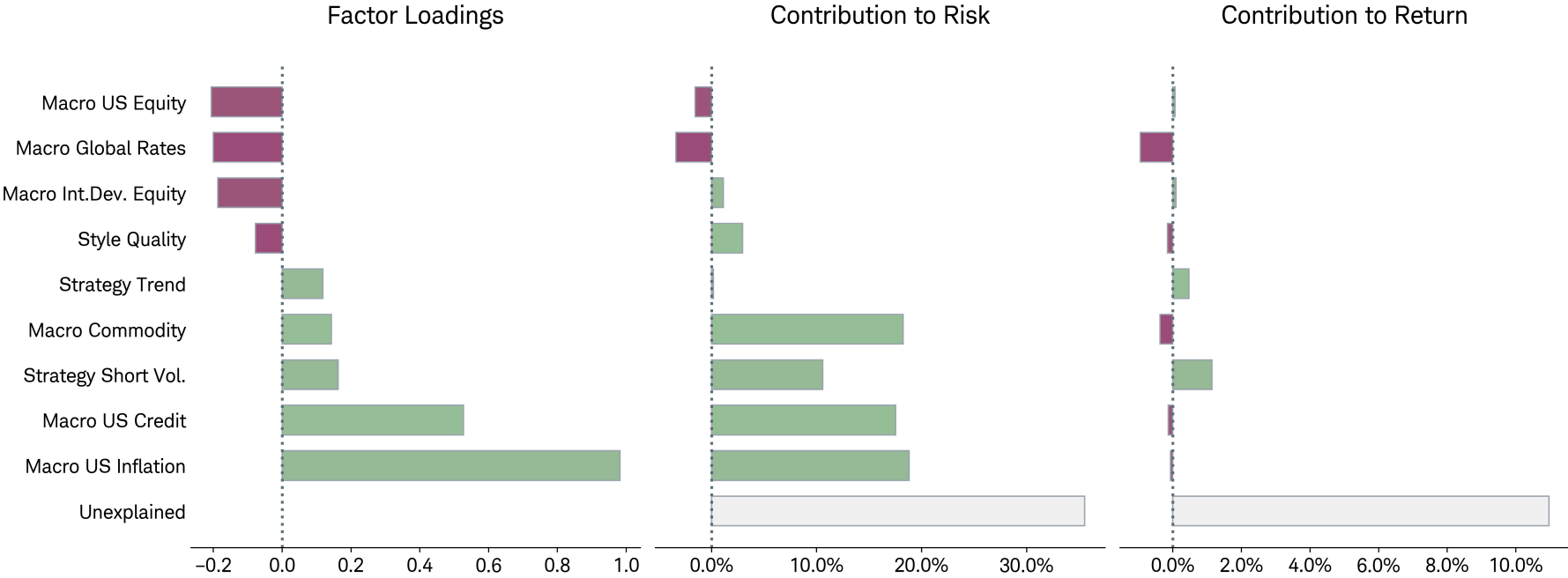
## Fund B Analytics

# Full History Factor Loadings and Related Beta Metrics

|                  |                 |             | Beta Estimate         | Contribution Risk | Contribution Return |
|------------------|-----------------|-------------|-----------------------|-------------------|---------------------|
| Selected Manager | Lookback Length | Return Date | Factor Name           |                   |                     |
| Fund B           | 279             | 03/31/2025  | Macro Global Rates    | -0.20             | -3.39%              |
|                  |                 |             | Macro Commodity       | 0.14              | 18.25%              |
|                  |                 |             | Macro US Inflation    | 0.98              | 18.81%              |
|                  |                 |             | Macro US Credit       | 0.53              | 17.53%              |
|                  |                 |             | Macro Int.Dev. Equity | -0.19             | 1.13%               |
|                  |                 |             | Macro US Equity       | -0.21             | -1.56%              |
|                  |                 |             | Style Quality         | -0.08             | 2.95%               |
|                  |                 |             | Strategy Short Vol.   | 0.16              | 10.59%              |
|                  |                 |             | Strategy Trend        | 0.12              | 0.18%               |
|                  |                 |             | Unexplained           |                   | 35.52%              |
|                  |                 |             | Sum                   |                   | 100.00%             |

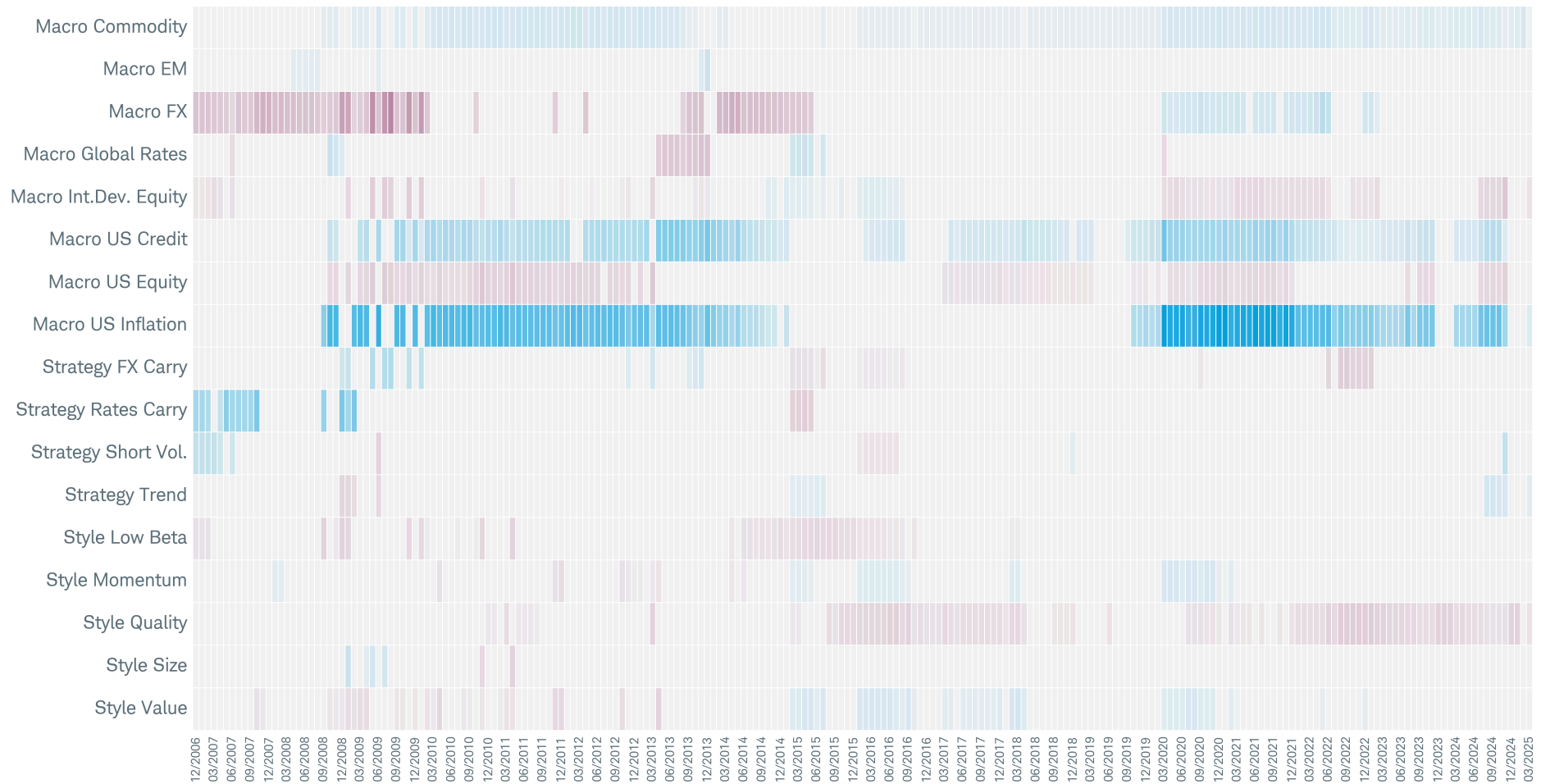
**Table 2.7** Summary of Fund B factor loadings, contribution to risk, and contribution to return. As computed from manager’s full history of 279 months ending 03/31/2025.

# Full History Factor Loadings and Related Beta Metrics

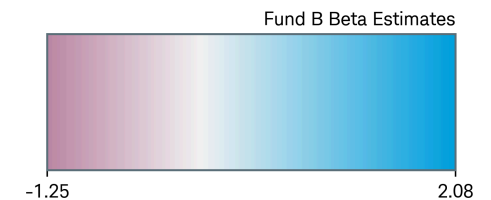


**Figure 2.11** Summary of Fund B factor loadings, contribution to risk, and contribution to return. As computed from manager's full history of 279 months ending 03/31/2025.

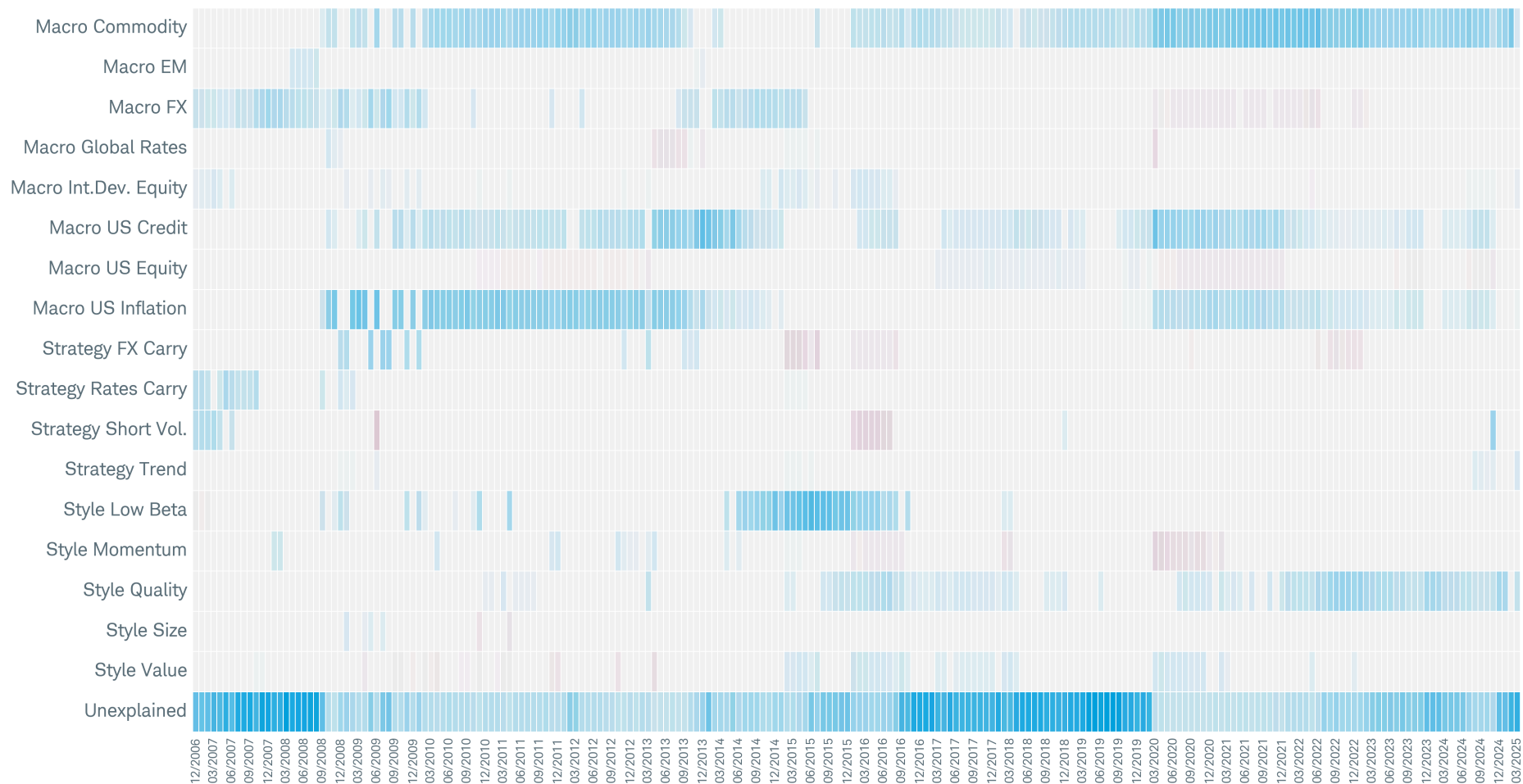
# Rolling Windows Factor Loadings (Beta Estimates)



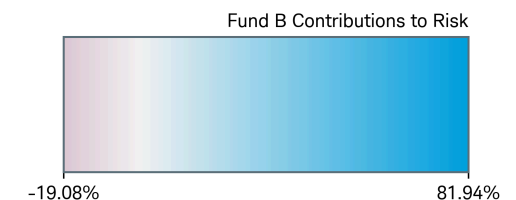
**Figure 2.12** Fund B heatmap of factor loadings (beta estimates) for 60-month rolling windows across manager's track record. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



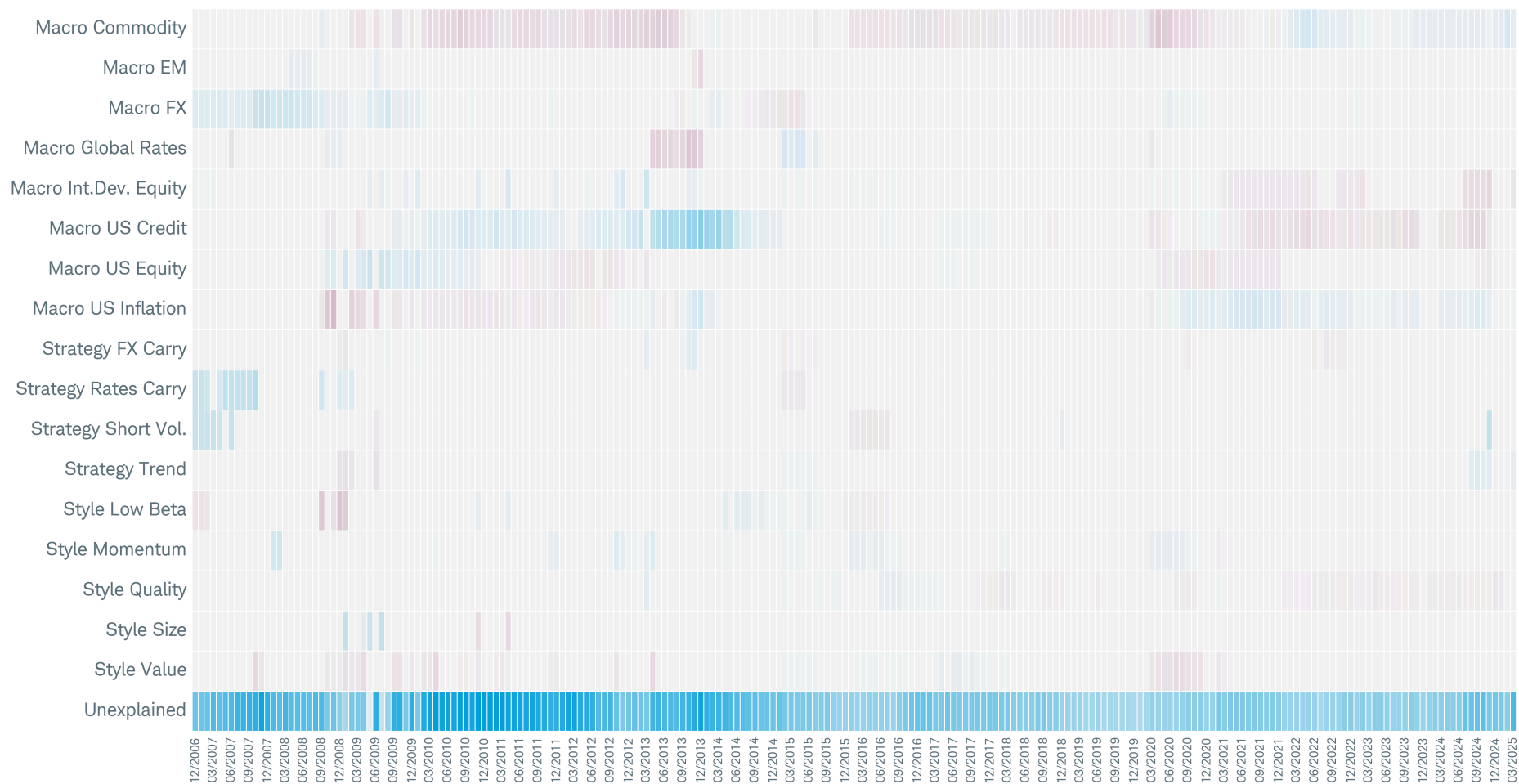
# Rolling Windows Contribution to Risk



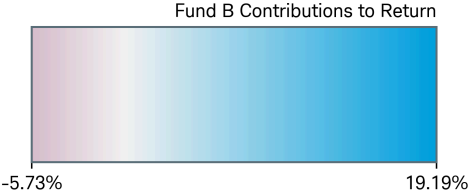
**Figure 2.13** Fund B heatmap of contribution to risk (derived from factor loadings) for 60-month rolling windows across manager's track record. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



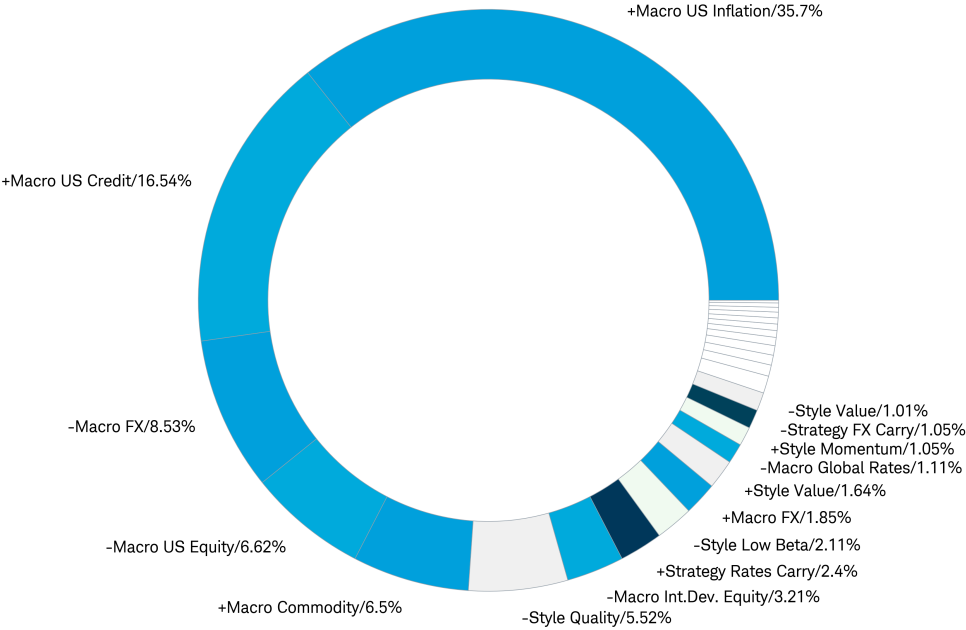
# Rolling Windows Contribution to Return



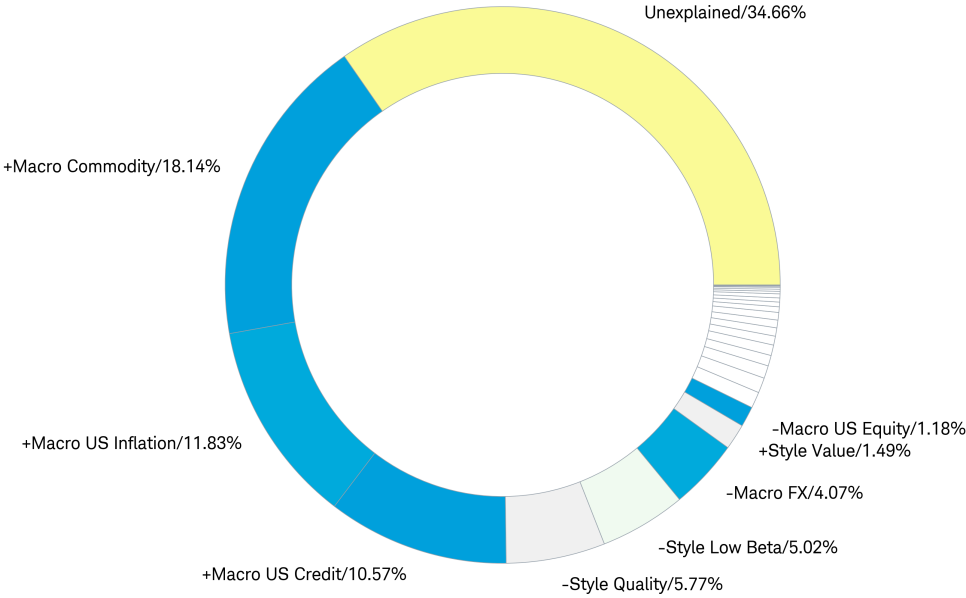
**Figure 2.14** Fund B heatmap of contribution to return (derived from factor loadings) for 60-month rolling windows across manager's track record. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



# Rolling Windows Factor Loadings and Contribution to Risk Summary



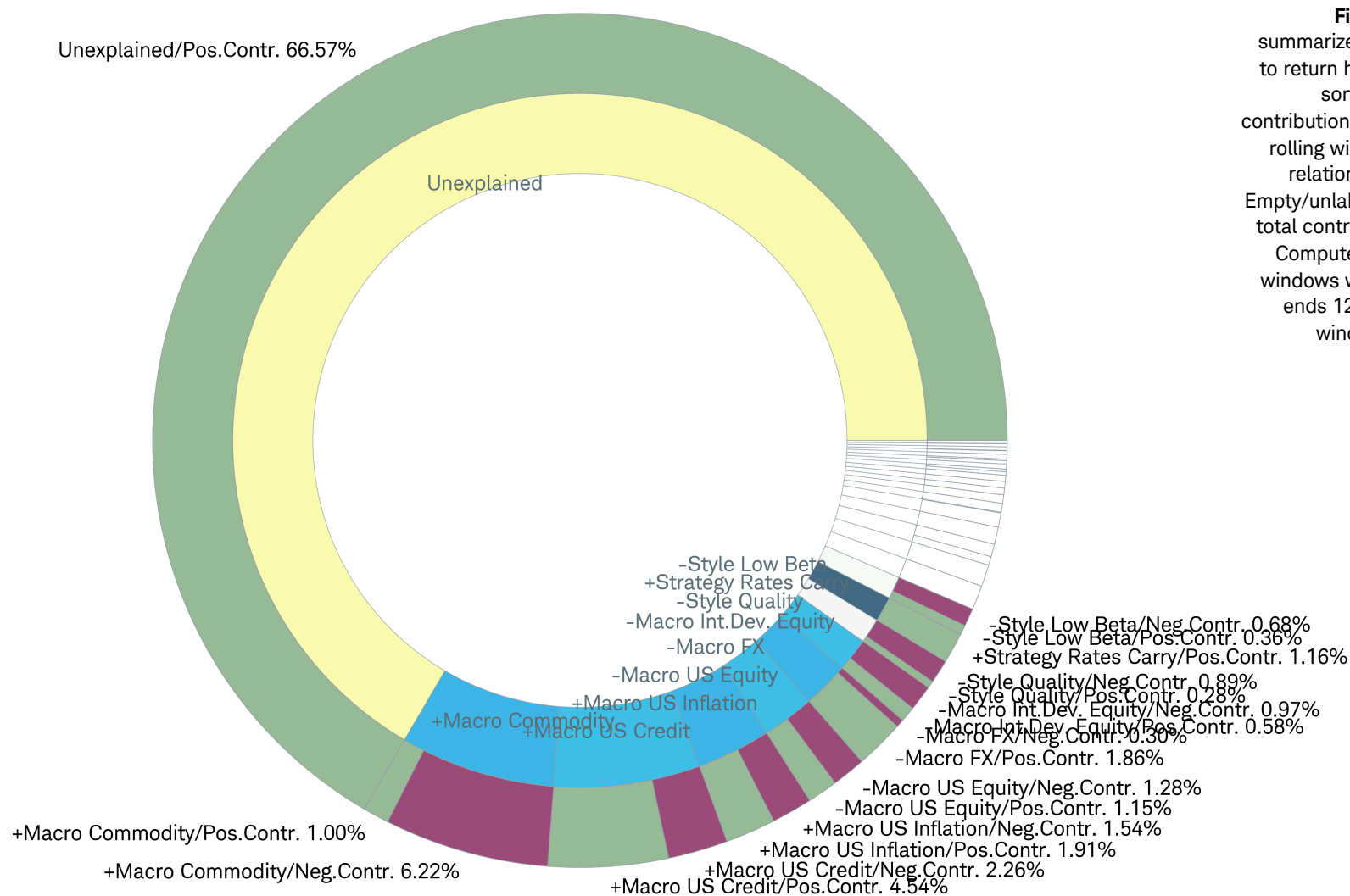
**Figure 2.15** Donut chart summarizes Fund B factor loading heatmap. Attribution is sorted by total percent of factor loadings across all rolling windows. Positive or negative beta indicated by +/- . Empty/unlabeled wedges indicate total loadings less than 1% of all loadings computed. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



**Figure 2.16** Donut chart summarizes Fund B contribution to risk heatmap. Attribution is sorted by percent of total risk computed across all rolling windows. Positive or negative prefix indicates contribution to risk associated with positive or negative beta. Empty/unlabeled wedges indicate a factor loading associated with less than 1% of all contribution to risk. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.



# Rolling Windows Contribution to Return Summary



# Rolling Windows Heatmap and Donut Summaries

|                     | Percent of Total<br>Factor Loadings | Percent of Loading<br>Positive Return | Percent of Loading<br>Negative Return | Percent of<br>All Return | Percent of Total<br>Risk |
|---------------------|-------------------------------------|---------------------------------------|---------------------------------------|--------------------------|--------------------------|
| Factor Loading      |                                     |                                       |                                       |                          |                          |
| Unexplained         |                                     | 66.57%                                | 0.00%                                 | 66.57%                   | 34.66%                   |
| +Macro Commodity    | 6.50%                               | 1.00%                                 | 6.22%                                 | 7.23%                    | 18.14%                   |
| +Macro US Inflation | 35.70%                              | 1.91%                                 | 1.54%                                 | 3.44%                    | 11.83%                   |
| +Macro US Credit    | 16.54%                              | 4.54%                                 | 2.26%                                 | 6.80%                    | 10.57%                   |
| -Style Quality      | 5.52%                               | 0.28%                                 | 0.89%                                 | 1.16%                    | 5.77%                    |
| -Style Low Beta     | 2.11%                               | 0.36%                                 | 0.68%                                 | 1.04%                    | 5.02%                    |
| -Macro FX           | 8.53%                               | 1.86%                                 | 0.30%                                 | 2.16%                    | 4.07%                    |
| +Style Value        | 1.64%                               | 0.33%                                 | 0.48%                                 | 0.81%                    | 1.49%                    |
| -Macro US Equity    | 6.62%                               | 1.15%                                 | 1.28%                                 | 2.43%                    | 1.18%                    |

**Table 2.8** Summary of Fund B factor loadings, contribution to risk, and contribution to return donut charts. Filtered for least 1% of risk. Percentages indicate the percent of all factor loadings, risk, and return computed across all rolling windows. Computed from 220 60-month windows where the first window ends 12/31/2006 and the last window ends 03/31/2025.

Beta is affected by both correlation and the ratio of standard deviations

$$\beta_{YX} = \rho_{YX} \frac{\sigma_Y}{\sigma_X}$$

And we also have correlation related to covariance by

$$\text{Cov}(X, Y) = \rho_{XY} \sigma_X \sigma_Y$$

If the dependent variable  $Y$  is much more volatile than the independent variable  $X$ , even with moderate correlation, beta can seem large. Correlation only measures linear association. Beta includes scale. *Note that pairwise betas are symmetric only when variables have the same mean and standard deviation as this ensures the scaling factors are identical in both directions.*

Pairwise Betas Unresidualized Factors Last 10 Years (07/31/2015 to 06/30/2025)

|                        |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
|------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
| Macro Commodity        | 1.00  | 0.42  | -1.27 | 0.29  | 0.79  | 1.28  | 0.5   | 3.49  | -0.41 | -0.53 | -1.02 | 1.01  | 0.69  | 1.02  | 0.59  | 1.03  | 0.59 |
| Macro Em               | 0.43  | 1.00  | -2.52 | 1.17  | 1.02  | 1.85  | 0.80  | 2.11  | -0.69 | -0.61 | -1.11 | 0.83  | 0.32  | 0.67  | 1.12  | 1.17  | -0.3 |
| Macro Fx               | -0.09 | -0.17 | 1.00  | -0.42 | -0.20 | -0.48 | -0.19 | -0.45 | 0.14  | 0.13  | 0.20  | -0.13 | -0.08 | 0.10  | -0.12 | -0.25 | 0.10 |
| Macro Global Rates     | 0.04  | 0.18  | -0.98 | 1.00  | 0.28  | 0.77  | 0.30  | 0.29  | -0.11 | -0.06 | -0.24 | 0.05  | -0.09 | -0.19 | 0.54  | 0.41  | -0.2 |
| Macro Int. Dev. Equity | 0.27  | 0.33  | -0.98 | 0.61  | 1.00  | 1.13  | 0.66  | 1.90  | -0.46 | -0.35 | -0.54 | 0.44  | 0.31  | 0.77  | 0.64  | 1.02  | 0.0  |
| Macro Us Credit        | 0.18  | 0.25  | -1.00 | 0.70  | 0.48  | 1.00  | 0.40  | 1.21  | -0.27 | -0.21 | -0.48 | 0.34  | 0.12  | 0.23  | 0.63  | 0.59  | -0.1 |
| Macro Us Equity        | 0.29  | 0.43  | -1.51 | 1.05  | 1.07  | 1.55  | 1.00  | 2.53  | -0.64 | -0.35 | -0.66 | 0.59  | 0.15  | 0.89  | 0.78  | 1.40  | -0.0 |
| Macro Us Inflation     | 0.10  | 0.06  | -0.19 | 0.05  | 0.16  | 0.25  | 0.13  | 1.00  | -0.08 | -0.08 | -0.13 | 0.14  | 0.10  | 0.30  | 0.21  | 0.22  | 0.0  |
| Style Low Beta         | -0.22 | -0.35 | 1.11  | -0.38 | -0.72 | -1.01 | -0.62 | -1.54 | 1.00  | 0.65  | 1.01  | -1.12 | -0.28 | -0.59 | 0.27  | -0.81 | 0.2  |
| Style Momentum         | -0.24 | -0.27 | 0.86  | -0.18 | -0.48 | -0.67 | -0.29 | -1.23 | 0.56  | 1.00  | 0.94  | -1.03 | -0.70 | -0.54 | -0.08 | -0.47 | 0.2  |
| Style Quality          | -0.15 | -0.16 | 0.44  | -0.23 | -0.24 | -0.50 | -0.18 | -0.68 | 0.28  | 0.31  | 1.00  | -0.58 | -0.15 | -0.30 | -0.18 | -0.32 | -0.0 |
| Style Size             | 0.17  | 0.13  | -0.32 | 0.05  | 0.21  | 0.39  | 0.18  | 0.79  | -0.35 | -0.37 | -0.65 | 1.00  | 0.30  | 0.20  | -0.03 | 0.29  | 0.0  |
| Style Value            | 0.24  | 0.11  | -0.40 | -0.20 | 0.32  | 0.30  | 0.09  | 1.16  | -0.18 | -0.54 | -0.34 | 0.63  | 1.00  | 0.29  | -0.09 | 0.27  | 0.0  |
| Strategy Fx Carry      | 0.08  | 0.05  | 0.12  | -0.10 | 0.19  | 0.14  | 0.13  | 0.85  | -0.09 | -0.10 | -0.17 | 0.10  | 0.07  | 1.00  | 0.36  | 0.22  | 0.0  |
| Strategy Rates Carry   | 0.01  | 0.02  | -0.04 | 0.08  | 0.04  | 0.10  | 0.03  | 0.16  | 0.01  | -0.00 | -0.03 | -0.00 | -0.01 | 0.10  | 1.00  | 0.06  | -0.0 |
| Strategy Short Vol     | 0.22  | 0.25  | -0.82 | 0.57  | 0.66  | 0.90  | 0.56  | 1.68  | -0.34 | -0.22 | -0.47 | 0.38  | 0.17  | 0.58  | 0.56  | 1.00  | 0.0  |
| Strategy Trend         | 0.07  | -0.04 | 0.19  | -0.17 | 0.01  | -0.15 | -0.00 | 0.22  | 0.06  | 0.06  | -0.07 | 0.03  | 0.03  | 0.11  | -0.19 | 0.05  | 1.00 |
| Macro Commodity        |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Em               |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Fx               |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Global Rates     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Int. Dev. Equity |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Us Credit        |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Us Equity        |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Macro Us Inflation     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Style Low Beta         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Style Momentum         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Style Quality          |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Style Size             |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Style Value            |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Strategy Fx Carry      |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Strategy Rates Carry   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Strategy Short Vol     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |
| Strategy Trend         |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |      |